Selection Purity Is Not a Sufficient Proxy for Transfer Utility: Auditing Joint Model and Data Heterogeneity in Language-Model Adaptation

Anonymous submission

Abstract

Federated language-model adaptation is often evaluated with homogeneous client architectures or task-consistent client data. We study these two forms of heterogeneity jointly: different decoder-only model families train on mixtures of useful, conflicting, and off-task examples. Our serialized protocols use real checkpoints, hashed manifests, clean-safety gates, official test sets and disjoint evaluation splits, and raw artifact hashes. On AG News, tokenizer-agnostic distillation from routed heterogeneous clients is 0.56 percentage points worse than a matched public-anchor-only server and costs $2.01\times$ the wall time. A post-observation check on the same test repeats the negative direction in all five new seeds (mean $-0.80\,pp$). On a separately frozen six-class TREC official evaluation, five seeds show no clear accuracy gain ($+0.40\,pp$, 95% seed interval $[-1.74, +2.54]\,pp$), while ECE falls by 0.0373 descriptively. After observing that result, a post-primary same-test mechanistic ablation with ten new seeds finds severe unfiltered transfer above clean transfer by $1.58\,pp$ (Holm $p = .015$), whereas the complete routed intervention is $1.42\,pp$ lower (Holm $p = .0074$) while lowering observed ECE. On GSM8K, an independently anchor-trained teacher used only to rank supplied answers improves a Qwen client by $3.34\,pp$ under severe contamination. Positive severe differences for SmolLM2 and TinyLlama fail a predeclared, post-observation Holm-corrected cross-family gate. Oracle-rate Co-Teaching is nonuniform on controlled and natural weak-label mixtures; both natural-noise development gates fail, so their test splits are not evaluated. Our evidence is diagnostic rather than algorithmic: routed transfer has no consistent accuracy benefit across two server tasks; both show lower observed ECE, a descriptive secondary pattern. In contrast, training-only selection helps one local Qwen setting without establishing beneficial federated transfer or cross-family stability.

Introduction

Federated fine-tuning promises to adapt language models without moving private client text. Two assumptions make this problem look easier than it is. First, many methods assume compatible client parameters even though deployed models can differ in width, tokenizer, and architecture. Second, evaluations often assume that a participating client's data is uniformly useful for the server task. Real client corpora can instead contain a mixture of useful examples, unrelated tasks, and responses that directly conflict with the target capability.

These axes interact. Parameter-space averaging may be undefined across model families, but moving to logits or probabilities only solves the interface problem; it does not establish whether transferred signals optimize the target server metric. Conversely, a local data filter that works for one model family may exploit architecture-specific loss scales and fail on another. We ask: *which conclusions survive when model-family and within-client data heterogeneity occur together?*

We make one evaluation contribution centered on the distinction between selection purity, probability quality, and downstream utility. We factor heterogeneous adaptation into local selection, canonical client-to-server transfer, and a public-anchor-only server control. Two finite-label server tasks instantiate this audit; open-answer and natural-weak-label studies test the boundary of the local-selection stage. We report conditional and negative outcomes with explicit pre-/post-observation scope, paired uncertainty, multiplicity control, cost, and content-addressed artifacts. This is a controlled diagnosis of where selection metrics and heterogeneous-transfer utility diverge, not a new optimizer or privacy mechanism. Figure 1 separates the server-transfer and local-selection branches.

Related Work

Federated PEFT and model heterogeneity. FedAvg aggregates compatible client parameters (McMahan et al. 2017), while FedMD and FedDF use public-data distillation to relax the shared-architecture assumption (Li and Wang 2019; Lin et al. 2020). LoRA makes adaptation economical by training low-rank updates (Hu et al. 2022), but factor exchange does not by itself resolve architectural incompatibility. FedBiOT compresses a server LLM for client-side adaptation (Wu et al. 2024). HetLoRA supports heterogeneous client LoRA ranks (Cho et al. 2024), while FLoRA avoids cross terms through stacking (Wang et al. 2024). FedMKT transfers knowledge between server LLMs and client SLMs (Fan et al. 2025). These works establish that heterogeneous LoRA and cross-size knowledge transfer are not new. We

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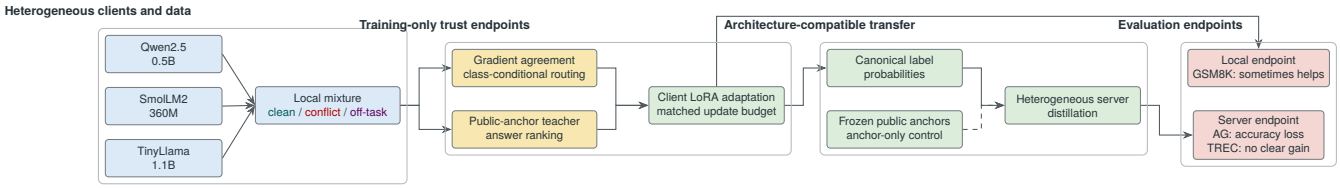


Figure 1: Evaluation framework. Heterogeneous client models train on mixtures of clean, conflicting, and off-task data. Routed client probabilities are transferred in the AG News and TREC server experiments; Co-Teaching and teacher ranking are local-adaptation controls. The public-anchor-only server is the matched transfer control, and local gains are not evidence of beneficial federated transfer.

test a distinct stress setting in which heterogeneous clients are also internally contaminated.

Noisy labels and data selection. The early-learning behavior of deep networks motivates small-loss selection (Arpit et al. 2017). Co-Teaching exchanges peer-selected small-loss examples (Han et al. 2018); trusted clean data can also estimate noise corrections (Hendrycks et al. 2018). Superfiltering shows that a weak language model can select instruction data for a stronger model (Li et al. 2024). Delora decouples a dual-LoRA noisy-label detector from downstream PEFT (Yuan, Chen, and Zhang 2025), and FedClean performs clean-sample selection, multi-stage finite-class relabeling, and robust aggregation (Jiang and Zhang 2025). Thus our public-anchor teacher score is a simple diagnostic endpoint. Delora and FedClean do not directly specify unbounded numeric relabeling or aggregation across incompatible model parameters; we report this interface incompatibility rather than claiming to outperform them.

Problem Setting

Clients $c \in \{1, \dots, C\}$ own decoder-only models f_c, tokenizers τ_c, and datasets D_c. The server model f_s may have a different architecture and tokenizer. A client dataset is a mixture

$$D_c = D_c^{\text{clean}} \cup D_c^{\text{conflict}} \cup D_c^{\text{off}},$$

where conflicting examples retain a target-task input but provide an incorrect answer, and off-task examples use a different task interface. Clients and server share a small public target-task anchor set A.

For architecture-agnostic classification exchange, client c transmits a probability vector over the canonical label set for every anchor input. The client score vector contains token-normalized candidate-answer losses $\boldsymbol{\ell}_{ci}$, giving $q_{ci} = \mathrm{softmax}(-\boldsymbol{\ell}_{ci})$ for anchor x_i. The server uses $\bar{q}_i = C^{-1} \sum_c q_{ci}$, computes its own canonical probabilities p_s, and minimizes $\mathcal{L}_s = \lambda_A \,\mathrm{CE}(y_i, p_s) + \lambda_D \,\mathrm{CE}(\bar{q}_i, p_s)$. The anchor-only control sets $\lambda_D = 0$; routed transfer uses the frozen λ_D. This interface is compatible by construction, but its utility depends on client data quality.

Training-Only Trust Endpoints

Class-conditional gradient routing

For a labeled example $z = (x, y)$, let $g(z)$ be the gradient of its causal answer loss with respect to local LoRA parameters and let a_y be the mean gradient of public anchors from class y. Their agreement is

$$s_g(z) = \frac{g(z)^\top a_y}{\|g(z)\| \|a_y\|}.$$

Let $P(v_i) \in [0, 1]$ denote the average-tie percentile rank of value v_i within client c, with lower values ranked first. For target-task examples, the inconsistency margin is $m(z) = \ell_y(z) - \min_k \ell_k(z)$; off-task examples receive one plus the largest observed target-task margin. The risk is

$$R(z) = \tfrac{1}{3}\{P(-s_g(z)) + P(\ell(z)) + P(m(z))\}.$$

Because raw cosine scales differ by model family, the routing gate uses the relative statistic

$$r_c = \frac{Q_{0.25}(\{s_g(z) : z \in D_c\})}{\max[Q_{0.25}(\{s_g(a) : a \in A\}), 10^{-12}]}.$$

The gate triggers when $r_c < 0.8$; otherwise all examples are retained. A triggered client keeps the lowest-risk $\max(1, \mathrm{round}(0.5|D_c|/K))$ examples per claimed target class, with source ID breaking ties; off-task examples have no claimed class and are excluded. Zero gradients are handled by the clamped cosine denominator. Thresholds and quotas were chosen on development data and fixed before the corresponding official-test runs. Clean-versus-conflict annotations are audit-only and never enter the score. Off-task membership is known from the explicit task interface; such examples have no claimed target class and are excluded whenever the gate triggers.

Public-anchor teacher ranking

Open numerical answers do not have a small canonical class set. We train an independent teacher f_T only on A and score a supplied pair $z = (x, \widetilde{y})$ by token-normalized answer loss,

$$s_T(z) = -\frac{1}{|\widetilde{y}|} \log p_{f_T}(\widetilde{y} \mid x).$$

After a numerical-interface check, the client retains the K lowest-loss examples and performs the same number of optimizer steps as the unfiltered baseline. No teacher parameter, hidden state, client validation label, or test example is used. The scalar endpoint is comparable across client model families, but requires an additional model and serial candidate scoring.

Experimental Protocol

Stage A: topic classification. AG News is the four-way target task and IMDb supplies real off-task text (Zhang, Zhao, and LeCun 2015; Maas et al. 2011). Conflicts use a deterministic non-identity label permutation. A frozen manifest separates 256 public anchors, 800 development examples, disjoint client pools, and the full 7,600-example official test. Clients are Qwen2.5-0.5B, SmolLM2-360M, and TinyLlama-1.1B; the server is Qwen2.5-1.5B.

Stage A2: independent question classification. TREC-Coarse provides a six-way answer-type task (Li and Roth 2002); IMDb remains the off-task source. Exact duplicate training lines are removed by raw-line hash before a balanced allocation of 72 anchors, 72 development questions, and three disjoint 120-question client pools. Each client class contributes 8 clean, 8 cyclic-conflict, and 4 off-task slots. The complete 500-question official test is evaluated over five seeds. The task, manifest, runner, full arguments, canonical $[72, 6]$ probability payloads, and serialized wire payloads are content-addressed. The model families, two rounds, update budget, and routing thresholds match Stage A. The parent primary protocol was frozen after instrumentation repairs and before its official evaluation; those repairs and invalidated smoke runs are recorded in its protocol history. The later four-arm protocol explicitly records prior test access and its own audit-counter repair before any retained full aggregate.

Stage B: numerical reasoning. GSM8K supplies target questions and SQuAD supplies off-task QA (Cobbe et al. 2021; Rajpurkar et al. 2016). Conflicts use deterministically rotated final numeric answers. The primary manifest separates public anchors, 512 development questions, and three client pools. Official evaluation uses all 1,319 test questions. After observing the primary phase, we froze a disjoint post-primary evaluation split of 1,024 training-split questions after excluding every anchor, development, and client-pool index; 4,913 unused questions remain. Answers are greedily generated, decimal-normalized, and scored by exact match. The client checkpoints are documented by their family reports (Qwen Team 2025; Ben Allal et al. 2025; Zhang et al. 2024).

Controls and statistics. Within each server task, comparisons match server initialization, public anchors, server optimizer steps, prompts, and seed. Anchor-only intentionally omits client training and communication, so end-to-end work is not matched and its wall-time advantage is reported. Local-selection comparisons match client optimizer steps. Each run-level protocol is serialized before its corresponding execution; this does not make later phases globally preregistered once an earlier phase has accessed the same benchmark test. We report paired differences and descriptive Student-t intervals over three seeds; these intervals characterize seed-to-seed variation, not task- or sample-level uncertainty. TREC uses a two-sided paired test over five designated official seeds because either direction is reportable; the directional improvement tests below are one-sided. Its post-primary ablation uses ten new common seeds and exactly three adjacent two-sided accuracy contrasts with one joint Holm correction; secondary metrics are descriptive and the already accessed test is disclosed. Declared families of primary contrasts use Holm correction. Raw JSON records carry evidence labels, model and manifest hashes, split scope, wall time, and peak GPU allocation; corpus-file hashes are present in the natural-label runs. All other reported improvement tests are one-sided paired Student-t tests over seed-matched differences. Holm correction is applied separately to the six AG News Co-Teaching contrasts, two cross-family GSM8K contrasts, nine family-by-severity secondary contrasts, and three contrasts within each natural-language development study.

Natural weak-label protocol. The COMPRISE Hausa and Yoruba topic datasets provide distant-supervision labels aligned with released clean training labels, plus separate clean development and test splits. Before training, we froze the measured disagreement rate as Co-Teaching's oracle forget rate, three new seeds per language, and a release gate requiring nonnegative mean gains for every family and at least one Holm-adjusted improvement. Clean training labels determine the single global oracle forget rate and support selection audits; per-example clean labels do not enter scoring, selection, or the training objective. A failed gate forbids test evaluation.

Metrics and optimization. For AG News and TREC, class scores are negative token-normalized answer-suffix losses; we report accuracy, macro-F1, the true-label suffix loss, and 10-bin ECE from the resulting canonical probabilities. GSM8K uses decimal-normalized exact match; natural-label studies use accuracy with macro-F1 as an imbalance audit. TREC additionally reports normalized canonical class NLL. All adapters target query/value projections with LoRA rank 4, scale 8, and no dropout, optimized by AdamW. The AG holdout uses two rounds of 40 client and 40 server steps (batch 8; server learning rate 10^{-4}); GSM8K uses 40 steps at batch 4 and 10^{-3}; natural-label runs use 256 steps at batch 8 and 10^{-3}. Frozen JSON protocols contain the remaining generation, length, and selection settings.

Results

Server distillation yields task-dependent accuracy utility

At development seed 42, severe AG News mixtures reduce local accuracy by 21.75–25.88 points across the three client families. The relative router is clean-safe in that three-family check; across Qwen seeds 42–44 it improves severe local accuracy by $+3.37$ to $+30.00\,pp$. This local improvement does not translate to the server endpoint.

The paired accuracy difference is $-0.56\,pp$, with descriptive 95% interval $[-0.76, -0.36]\,pp$; all seed-wise differences are negative. Although routed distillation lowers answer loss and 10-bin ECE, it costs $2.01\times$ the matched wall time. Both accuracy and macro-F1 decline; no joint accuracy–macro-F1 decision threshold was predeclared. Thus adding routed client probability targets is harmful under AG News task accuracy in this pipeline, not uniformly worse across predictive metrics. After observing this result, we froze five additional seeds without retuning. All five routed-minus-anchor differences are negative, with mean $-0.80\,pp$

Table 1: Server results. Accuracy is mean ± SD; Macro-F1 and ECE are seed means. Rows below the rule are the post-primary reused-test TREC ablation with ten new seeds.

Task	Method	Accuracy	Macro-F1	ECE
AG News	Anchor-only	84.34 ± 0.62	84.28	0.054
AG News	Routed	83.78 ± 0.54	83.67	0.016
TREC	Anchor-only	80.72 ± 1.41	77.65	0.068
TREC	Routed	81.12 ± 0.67	78.38	0.031
TREC post	Anchor-only	81.76 ± 2.01	78.39	0.060
TREC post	Clean uniform	81.56 ± 1.55	79.01	0.035
TREC post	Severe uniform	83.14 ± 0.77	80.80	0.099
TREC post	Severe routed	81.72 ± 1.38	79.43	0.039

and descriptive interval $[-1.18, -0.42]\,pp$. The predeclared 4-of-5 directional seed-consistency heuristic passes, but it is not an alpha-controlled test and reuses the already accessed official test. The combined eight-seed estimate ($-0.71\,pp$) is descriptive and does not replace the original three-seed analysis.

TREC changes the accuracy conclusion but not the calibration pattern. Across five seeds, routed-minus-anchor accuracy ranges from -1.8 to $+2.6\,pp$ with mixed signs. The mean is $+0.40\,pp$, with descriptive 95% seed interval $[-1.74, +2.54]\,pp$ and two-sided paired $p = 0.631$. Macro-F1 changes by $+0.73\,pp$ with an interval that also crosses zero. Canonical class NLL decreases by 0.0304 ($p = 0.0616$), while the secondary ECE decreases by 0.0373 in every seed, with interval $[-0.0487, -0.0259]$ and nominal, multiplicity-unadjusted $p = 0.00081$. Thus TREC does not validate an accuracy benefit; the interval spans modest harm and benefit. Its lower ECE is a descriptive secondary pattern. Routing triggers in 27/30 client-rounds and records 19,830 bytes of serialized probability tensors per run, excluding transport metadata. Its $9.61\times$ mean timed-region per-process wall-time ratio was measured with anchor runs serial and routed runs up to two-way concurrent; it is neither end-to-end nor an equal-work deployment benchmark.

After observing this compound comparison, we froze a post-primary four-arm ablation on the already accessed TREC test with ten new common seeds. The three adjacent accuracy contrasts form one two-sided Holm family. Clean uniform transfer versus anchor-only is inconclusive ($-0.20\,pp$, interval $[-0.71, +0.31]$, Holm $p = .397$). Contrary to a simple contamination-harm story, severe unfiltered transfer exceeds clean transfer by $1.58\,pp$ ($[+0.54, +2.62]$, Holm $p = .015$). The complete routed intervention then reduces severe-uniform accuracy by $1.42\,pp$ ($[-2.19, -0.65]$, Holm $p = .0074$), while descriptively reducing class NLL by 0.0314 and ECE by 0.0596. The routed-minus-anchor net is $-0.04\,pp$ with interval $[-0.84, +0.76]$, without a fourth test. This same-test analysis diagnoses an accuracy–calibration trade-off; it is not a new holdout or independent confirmation and does not isolate which mixture component produces the severe-uniform gain. Routing triggers in 57/60 client-rounds and raises the selected clean fraction from 40.0% to 64.6% when off-task rows remain in the denominator, yet lowers accuracy relative to severe uniform transfer. Selection purity is therefore not a sufficient proxy for server utility.

Table 2: AG News development accuracy over three seeds. Parentheses are Co-Teaching-minus-naive gains; six contaminated tests are jointly corrected.

Client	Conflict 40%	Joint 40%/20%
Qwen	59.12 / 60.92 (+1.79)	53.71 / 54.50 (+0.79)
SmolLM2	65.83 / 68.46 (+2.63)	48.00 / 47.37 (-0.63)
TinyLlama	59.25 / 68.54 (+9.29)	54.92 / 37.25 (-17.67)

Oracle-rate Co-Teaching is not uniformly robust

We compare naive LoRA to two-peer Co-Teaching on the same frozen AG News development mixtures. Co-Teaching is given the true total contamination fraction as its forget rate, an oracle advantage, and its peer probabilities are ensembled at evaluation. Table 2 shows that this strong compatible baseline is highly variable: it helps TinyLlama under label conflict but sharply hurts it when off-task data is added.

No contrast survives Holm correction (all adjusted $p = 1.0$); seed-wise effects change sign in five of six settings. Co-Teaching uses two adapters and costs 2.84–$9.32\times$ the naive training wall time in the concurrently scheduled diagnostic runs; these are environment-specific ratios, not isolated deployment benchmarks. It therefore does not explain away the benchmark failure with an established small-loss baseline, although the joint off-task condition exceeds its original finite-label-noise scope.

Natural weak labels do not yield uniform robustness

We next use Hausa and Yoruba news-topic data from a multilingual weak-supervision study (Hedderich et al. 2020). Its distant-supervision labels disagree with the released clean labels on 50.37% of 2,045 training headlines. We froze this rate as Co-Teaching's oracle forget rate, trained on all weak labels, and used only the 290-example clean development split for evaluation. Across three seeds, naive/Co-Teaching accuracies are 44.94/33.33 for Qwen, 45.29/28.16 for SmolLM2, and 47.82/28.74 for TinyLlama: changes of -11.61, -17.13, and $-19.08\,pp$, respectively. The predeclared development gate fails (all three Holm-adjusted $p = 1.0$); the clean test split is therefore not evaluated. In the concurrently scheduled diagnostic runs, Co-Teaching costs 1.69–$1.80\times$ the naive wall time and two adapters; these ratios are not isolated deployment benchmarks. The separately frozen Yoruba development matrix has 33.28% disagreement over 1,340 headlines. Its Co-Teaching changes are $+0.53$, $+2.65$, and $-4.23\,pp$ for Qwen, SmolLM2, and TinyLlama, respectively; all intervals cross zero and no improvement test survives Holm correction. This second gate also fails, so its clean test split is not evaluated. The Yoruba estimates are closer to zero, but this cross-language comparison does not isolate the effect of noise rate. Figure 2 summarizes these paired natural-label effects and their seed-level uncertainty.

Table 3: Non-Qwen GSM8K extension, frozen before those runs but after the Qwen phase had accessed the same official test. Holm p covers the two declared severe hypotheses.

Client	Condition	Naive	Teacher	Gain
SmolLM2	clean	4.55	4.90	+0.35
SmolLM2	severe	2.58	4.17	+1.59
TinyLlama	clean	3.66	4.25	+0.58
TinyLlama	severe	1.42	3.44	+2.02

Higher selection purity is not sufficient for improved accuracy. Hausa weak labels match the released clean labels on 49.63% of training examples, while the examples selected by Co-Teaching match clean labels on only 45.3%–50.9% across families: small loss provides no consistent enrichment. On Yoruba, selection raises clean-label agreement from 66.72% in the training population to 78.2%–80.6%. Nevertheless, its accuracy effects remain mixed. Thus the Hausa pattern is consistent with selection collapse, but selection purity alone does not guarantee beneficial adaptation.

GSM8K teacher ranking helps Qwen

On the official GSM8K test, Qwen naive severe adaptation reaches 3.49% EM and teacher-ranked adaptation reaches 6.82%, a $+3.34\,pp$ paired mean. Clean EM increases from 6.62% to 7.91%. This supports the narrow conclusion that an independent training-only teacher signal can make supplied numerical answers more useful in this Qwen setting.

Post-observation cross-family effects do not pass their gate

All six severe paired differences are positive and both clean safety gates hold. Nevertheless, both severe intervals cross zero and both Holm-adjusted tests have $p = 0.0714$. The declared general cross-family claim fails.

Matched controls and severity curve

On development data, teacher ranking has higher severe mean EM than both a numeric-random 128-example selector and an unadapted-client small-loss selector. The gains over these controls are $+0.78/+0.98\,pp$ for SmolLM2 and $+1.56/+1.50\,pp$ for TinyLlama. These post-hoc controls support a teacher-signal interpretation but cannot repair the failed holdout gate.

The frozen descriptive curve gate passes: every clean seed remains above the $-2\,pp$ safety bound and every family has nonnegative mean gain at medium and severe contamination. None of nine family-by-severity tests survives joint Holm correction, so Table 4 is trend evidence rather than nine confirmatory effects.

Cost

In a concurrently executed, matched 64-question diagnostic, teacher ranking costs $1.52\times$ the naive SmolLM2 wall time and $1.58\times$ the naive TinyLlama time. Serial scoring of 256 candidates takes approximately 8.1 seconds and the teacher path peaks at 5.57 GiB allocated GPU memory; these are environment- specific diagnostics, not isolated throughput measurements.

Discussion

Across the two server tasks, canonical-probability exchange is compatible by construction but has no consistent accuracy benefit: the tested pipeline is harmful on AG News, while the TREC effect is inconclusive and its interval spans modest harm and benefit. Both tasks show lower observed 10-bin ECE, a secondary pattern that motivates a dedicated calibration study. The post-primary TREC decomposition further rejects a simple explanation in which contamination makes client probabilities harmful and routing repairs them: severe uniform transfer has the highest accuracy, whereas routing trades away that gain while lowering descriptive NLL and ECE. Function-space compatibility is therefore necessary but insufficient for accuracy gains. The Qwen result is consistent with the hypothesis that a clean training-only reference ranks conflicting supplied answers better than the unadapted client's own loss; the present experiments do not establish this mechanism across families. Cross-family estimates point in the same direction but are too uncertain to support a universal claim.

Three distinct failure modes. First, LoRA factors from different architectures lack a shared parameter coordinate system; function-space exchange resolves this interface failure. Second, canonical probabilities can encode task- and metric-dependent utility: the same transfer pipeline can lose accuracy, gain accuracy, or improve only calibration depending on its selection policy. Third, small-loss or teacher-based selection is distribution- and model-dependent: Hausa selection collapses, whereas Yoruba purity improves without reliable accuracy gains. Conflating these failures can make an aggregation interface appear robust while leaving knowledge utility untested.

Implication. A robust audit should compare an anchor-only server with clean uniform, contaminated uniform, and routed contaminated arms. It should distinguish server-side from end-to-end optimizer work and audit selected composition. Otherwise, a near-zero net contrast can conceal large opposing mixture and selection effects, and better calibration can be mistaken for accuracy utility.

Overall, the evidence supports an evaluation contribution rather than a new federated optimizer. AG News diagnoses a server-accuracy failure and TREC shows both an inconclusive net effect and a post-primary selection-induced accuracy–calibration reversal, while the controlled and natural-label studies show that established local-selection baselines are not uniformly robust. These findings are complementary but concern distinct endpoints; they do not turn the training-only teacher endpoint into a federated algorithm.

Limitations and Ethics

Conflicts are controlled synthetic interventions and do not represent all annotation processes. The Hausa and Yoruba studies add naturally occurring weak classification labels but are development-only; naturally occurring numerical label

Table 4: Disjoint 1,024-example post-primary severity curve, frozen after the primary result and used descriptively. Values are mean EM; parentheses show teacher-minus-naive percentage-point gain.

	Clean		Medium (20/10)		Severe (40/20)	
Client	Naive	Teacher	Naive	Teacher	Naive	Teacher
Qwen	6.38	6.84 (+0.46)	5.24	7.85 (+2.60)	1.69	5.76 (+4.07)
SmolLM2	3.68	2.80 (-0.88)	2.15	3.39 (+1.24)	1.46	2.47 (+1.01)
TinyLlama	2.99	2.93 (-0.07)	2.02	2.44 (+0.42)	1.07	1.73 (+0.65)

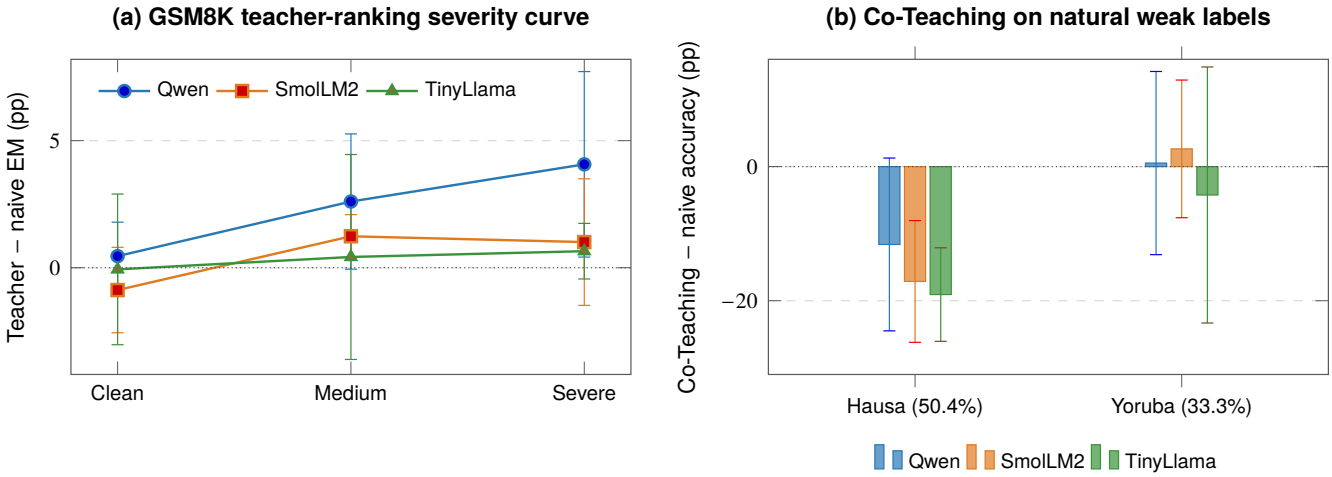


Figure 2: Paired mean changes with descriptive 95% Student-t intervals over three seeds; intervals describe seed variation only. (a) Teacher ranking trends upward with GSM8K contamination, but none of nine jointly corrected contrasts is significant. (b) Oracle-rate Co-Teaching is uniformly harmful on higher-noise Hausa and mixed on lower-noise Yoruba; both frozen development gates fail and neither clean test split is evaluated.

errors remain untested. Most comparisons use three seeds, giving unstable intervals and low power after multiplicity correction. The post-observation five-seed AG check strengthens directional consistency but does not add a new task or untouched test set. The public-anchor teacher adds compute and may encode biases from pretraining. TREC adds an independently sourced task and five designated official seeds, but its final protocol followed documented instrumentation repairs, its official test contains only 500 questions, its published license is recorded as unknown, and language-model pretraining contamination cannot be ruled out. The ten-seed decomposition was designed after that test and its primary aggregate had been observed; corrected contrasts on the reused test are mechanistic, not independent confirmation. The severe arm jointly changes conflict and off-task content, so its $1.58\,pp$ effect cannot be assigned to either component or generalized to natural corruption. Neither logits, losses, nor anchors provide formal privacy; membership or reconstruction leakage is not measured. We make no privacy-preserving, production, or broad generalization claim. FedClean and Delora are not directly applicable to the open-answer heterogeneous interface, but we have not shown that our endpoint outperforms them in their native homogeneous classification setting.

AI-assistance disclosure. OpenAI Codex assisted with code drafting, experiment orchestration, artifact validation, and manuscript editing. The human authors are responsible for the study design and submission, verified the cited sources and reported results against the raw artifacts, and take full responsibility for the paper's content.

Reproducibility

The artifact includes hashed manifests, serialized JSON run protocols, exact model weight hashes, per-run raw records, deterministic mixture construction, generated Markdown summaries, and unit tests. The official-test and secondary-holdout records are explicitly distinguished. The paper tables are derived from source records rather than manually entered experiment logs. Because these artifacts are currently workspace files rather than an external timestamped preregistration, their hashes establish internal consistency, not independent proof of creation time. A sidecar clarification records the natural-noise oracle-rate use without rewriting its original protocol. The primary TREC schema embeds protocol, runner, task-core, model, data, split, argument, mixture, simplex, and wire-payload hashes and is checked as a full 5-by-2 matrix. The post-primary TREC artifact separately checks a complete 10-by-4 matrix, three frozen adjacent contrasts, invalidated instrumentation history, and final summarizer/verifier hashes. The legacy AG/GSM record schema does not embed the protocol digest, complete CLI, or ev-

ery evaluation-file hash; current verifiers therefore establish artifact consistency but not independent historical provenance. The runtime exposed one device labeled NVIDIA GeForce RTX 4090 with 49,140 MiB visible memory; this host-reported capacity is not a claim about retail-device specifications. Experiments used Linux 6.8 with Python 3.10.20, PyTorch 2.11.0+cu128, Transformers 4.57.6, PEFT 0.18.1, and SciPy 1.15.3. Primary comparisons use three explicitly recorded seeds; primary TREC uses five seeds, its disclosed post-primary ablation ten new seeds, and the AG post-observation check five separately reported seeds. Protocol files contain the final optimizer, batch, LoRA, and generation settings.

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